That is, when there is no reaction, the concentration and momentum boundary-layer thicknesses bear a constant ratio to one another, dependent only on the value of the Schmidt number.

When the *reaction is slow*
be obtained:

$\Delta =$

in which $\xi = [12/(n+1)](k$
19.2–35 gives $a_1 = -(\frac{1}{7})$Sc
boundary-layer thickness is

from BSL - Transport Phenomena 1960

§19.3 BOUNDARY-LAYER THEORY: EXACT SOLUTIONS FOR SIMULTANEOUS HEAT, MASS, AND MOMENTUM TRANSFER

The boundary-layer developments we have discussed thus far are based on the von Kármán integral forms of the boundary-layer equations (Eqs. 4.4–29, 11.4–23, and 19.2–25), which are obtained from the Prandtl boundary-layer equations by integrating in the y-direction. To solve the von Kármán equations for the boundary-layer thicknesses δ, δ_T, and δ_c, one has to assume the shapes of the velocity, temperature, and concentration profiles. In this section we illustrate the more rigorous procedure of solving the Prandtl boundary-layer equations exactly;[1] this method gives greater accuracy and better illustrates the details of the transport mechanisms.

We are interested here in determining what happens when heat, mass, and momentum are simultaneously transferred across the boundary of a flowing fluid. This behavior is important in connection with many processes, such as

[8] See P. L. Chambré and J. D. Young, *Physics of Fluids*, **1**, 48–54 (1958). Catalytic surface reactions in boundary layers have been studied by P. L. Chambré and A. Acrivos, *J. Appl. Phys.*, **27**, 1322–1328 (1956).

[1] In the boundary-layer literature the term "exact solution" denotes a solution of the Prandtl boundary-layer equations as opposed to more approximate solutions. It should be noted that the Prandtl boundary-layer equations are actually asymptotic approximations to the equations of change and are accurate for two-dimensional laminar flow in the region $v_\infty x/\nu \gg 1$. (See H. Schlichting, *Boundary-Layer Theory*, Pergamon Press, New York (1955), Chapter VII.)

combustion of solid fuels, heterogeneous catalysis, and separation processes. The system chosen here is a simple one, but a study of the results can give valuable insight regarding the behavior of more complex systems.

Consider the flow system shown in Fig. 19.3–1. A thin, semi-infinite plate of volatile solid A sublimes, under steady conditions, into an unbounded gaseous stream of A and B, which approaches the plate tangentially in the x-direction with velocity v_∞. Species B is present in the gas phase only. We wish to determine the profiles of velocity, temperature, and concentration in the boundary layer for the case of known, uniform temperature and gas composition along the plate surface.

For simplicity, we assume that there are no chemical reactions and no external forces other than gravity, and we neglect viscous dissipation, heats of mixing, and emission and absorption of radiant energy in the gaseous boundary layer. These assumptions are reasonable for many gaseous systems. In addition, we treat the physical properties ρ, μ, $\hat{C}_p$, k, c, and $\mathscr{D}_{AB}$ as constants.[2] With these simplifications, the boundary-layer equations for this system become

(continuity)
$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0 \tag{19.3–1}$$

(motion)
$$v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = \nu \frac{\partial^2 v_x}{\partial y^2} \tag{19.3–2}$$

(energy)
$$v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2} \tag{19.3–3}$$

(continuity of A)
$$v_x \frac{\partial x_A}{\partial x} + v_y \frac{\partial x_A}{\partial y} = \mathscr{D}_{AB} \frac{\partial^2 x_A}{\partial y^2} \tag{19.3–4}$$

The boundary conditions are

$$\text{at } x \leqslant 0 \text{ or } y = \infty: \quad v_x = v_\infty, \quad T = T_\infty, \quad x_A = x_{A\infty} \tag{19.3–5,6,7}$$

$$\text{at } y = 0: \quad v_x = 0, \quad T = T_0, \quad x_A = x_{A0}, \quad N_B = 0 \tag{19.3–8,9,10,11}$$

The last equation follows from the assumptions that B exists only in the gas phase and that there are no chemical reactions.

The equation of continuity may be integrated to obtain v_y as follows:

$$v_y = v_{y0} - \int_0^y \frac{\partial v_x}{\partial x}\,dy \tag{19.3–12}$$

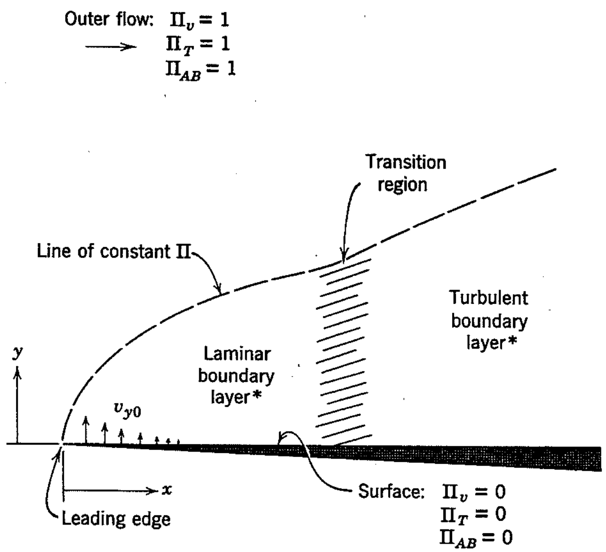
or

$$v_y = \frac{M_A N_{A0}}{\rho} - \int_0^y \frac{\partial v_x}{\partial x}\,dy \tag{19.3–13}$$

[2] Note that these assumptions imply that $M_A = M_B$ and $\hat{C}_{pA} = \hat{C}_{pB}$, inasmuch as x_A is a function of position. Approximate methods of applying the present solutions to systems with variable properties are discussed in §21.7. Variation in the physical properties can be treated accurately by the numerical integration method of H. Schuh, *Z. angew. Math. Mech.* **25–27**, 54–60 (1947); *NACA Tech Memo 1275* (1950).

in which the subscript 0 denotes conditions at $y = 0$. The second relation follows from Eqs. J and S in Table 16.1–3. Evaluation of N_{A0} from Fick's law (Eq. 16.2–2) gives

$$N_{A0} = -c\mathscr{D}_{AB} \left.\frac{\partial x_A}{\partial y}\right|_{y=0} + x_{A0}(N_{A0} + N_{B0}) \tag{19.3–14}$$



*The boundary layer below the plate is omitted here.

Fig. 19.3–1. Tangential flow along a sharp-edged semi-infinite flat plate with mass transfer into the stream. The laminar-turbulent transition usually occurs at a *length* Reynolds number ($v_\infty x/\nu$) on the order of 10^5.

By setting $N_{B0} = 0$ from Eq. 19.3–11 and combining the result with Eq. 19.3–13, we get

$$v_y = -\frac{M_A}{\rho}\frac{c\mathscr{D}_{AB}}{(1 - x_{A0})}\left.\frac{\partial x_A}{\partial y}\right|_{y=0} - \int_0^y \frac{\partial v_x}{\partial x}\,dy \tag{19.3–15}$$

This expression is to be used for v_y in Eqs. 19.3–2, 3, and 4. Because we have assumed that $M_A = M_B$, we may set $M_A c/\rho$ equal to unity.

We now define the dimensionless profile variables:

$$\Pi_v = \frac{v_x - v_{x0}}{v_{x\infty} - v_{x0}} = \frac{v_x}{v_\infty}; \quad \Pi_T = \frac{T - T_0}{T_\infty - T_0}; \quad \Pi_{AB} = \frac{x_A - x_{A0}}{x_{A\infty} - x_{A0}} \tag{19.3–16,17,18}$$

and the dimensionless physical property ratios

$$\Lambda_v = \frac{\nu}{\nu} = 1; \quad \Lambda_T = \frac{\nu}{\alpha} = \mathrm{Pr}; \quad \Lambda_{AB} = \frac{\nu}{\mathscr{D}_{AB}} = \mathrm{Sc} \tag{19.3–19,20,21}$$

With these definitions, the equations of motion, energy, and continuity of A take the common form

$$v_x\frac{\partial \Pi}{\partial x} - \left(\frac{\mathscr{D}_{AB}}{(1 - x_{A0})}\left.\frac{\partial x_A}{\partial y}\right|_{y=0} + \int_0^y \frac{\partial v_x}{\partial x}\,dy\right)\frac{\partial \Pi}{\partial y} = \frac{\nu}{\Lambda}\frac{\partial^2 \Pi}{\partial y^2} \tag{19.3–22}$$

and the boundary conditions on v_x, T, and x_A become

$$\text{at } x \leqslant 0 \quad or \quad y = \infty, \qquad \Pi = 1 \tag{19.3–23}$$

$$\text{at } y = 0, \qquad \Pi = 0 \tag{19.3–24}$$

The form of the boundary conditions on the profile variables Π, and the lack of a characteristic length in the flow system, suggest that the method of combination of variables may be used. Equation 4.4–26 suggests a combination of the form

$$\eta = \frac{y}{2}\sqrt{\frac{v_\infty}{\nu x}} \tag{19.3–25}$$

By expressing the derivatives in Eq. 19.3–22 in terms of η, we obtain[3]

$$\Lambda\left[\frac{1}{\Lambda_{AB}}\left(\frac{x_{A0} - x_{A\infty}}{1 - x_{A0}}\right)\Pi'_{AB}(0) - \int_0^\eta 2\Pi_v\,d\eta\right]\Pi' = \Pi'' \tag{19.3–26}$$

in which primes denote differentiation with respect to η. We see from this expression that the rate of mass transfer at the wall, expressed here in terms of the concentration gradient $\Pi'_{AB}(0)$, directly affects the three profiles Π_v, Π_T, and Π_{AB}.

For later convenience, we define the quantity

$$f = -K + \int_0^\eta 2\Pi_v\,d\eta \tag{19.3–27}$$

The quantity K is a dimensionless mass flux at the wall, given here by

$$K = \frac{1}{\Lambda_{AB}}\left(\frac{x_{A0} - x_{A\infty}}{1 - x_{A0}}\right)\Pi'_{AB}(0) \tag{19.3–28}$$

[3] It is instructive to compare the development at this point with that given in Example 19.1–1. The analogous quantities in the two developments are for small y:

Example 19.1–1	$1 - X$	Z	$-\left.\frac{dX}{dZ}\right\|_{Z=0}$	2φ	$\frac{x_{A0}}{1 - x_{A0}}$
This section	Π_{AB}	η	$\Pi'_{AB}(0)$	$K\Lambda_{AB}$	$\frac{x_{A0} - x_{A\infty}}{1 - x_{A0}}$

and is constant for the laminar boundary layer.[4] The boundary-layer equations of motion, energy, and continuity then become

$$-\Lambda f \Pi' = \Pi'' \tag{19.3-29}$$

The boundary conditions on Π are

$$\text{at } \eta = 0, \qquad \Pi = 0 \tag{19.3-30}$$

$$\text{at } \eta = \infty, \qquad \Pi = 1 \tag{19.3-31}$$

The given differential equations and their boundary conditions have now been expressed in terms of the single independent variable η, confirming the assumed combination of variables.

From Eqs. 19.3-27 through 31, we can compute the profiles of velocity, temperature, and concentration in the laminar boundary layer. We first compute the velocity profiles Π_v and the related function f for the desired value of K by solving Eq. 19.3-29 with $\Lambda = 1$; then the same equation can be directly integrated for any value of Λ to obtain the corresponding temperature or concentration profile:

$$\Pi(\eta, \Lambda, K) = \frac{\displaystyle\int_0^{\eta} \exp\left(-\Lambda \int_0^{\eta} f\, d\eta\right) d\eta}{\displaystyle\int_0^{\infty} \exp\left(-\Lambda \int_0^{\eta} f\, d\eta\right) d\eta} \tag{19.3-32}$$

Some profiles computed numerically from Eq. 19.3-32 are given in Fig. 19.3-2. Velocity profiles are given by the curves for $\Lambda = 1$; temperature and concentration profiles for various Prandtl and Schmidt numbers are given by the curves for the corresponding values of Λ. Note that the velocity, temperature, and concentration boundary layers get thicker when the dimensionless mass-transfer rate K is positive (as in evaporation) and thinner when K is negative (as in condensation).

The dimensionless gradients of velocity, temperature, and concentration at the wall, $\Pi'(0, \Lambda, K)$, are obtained from the derivative of Eq. 19.3-32:

$$\Pi'(0, \Lambda, K) = \frac{1}{\displaystyle\int_0^{\infty} \exp\left(-\Lambda \int_0^{\eta} f\, d\eta\right) d\eta} \tag{19.3-33}$$

A summary of values computed from this formula by numerical integration is given in Table 19.3-1.

[4] The dimensionless mass flux K is defined as

$$K = \frac{2v_{y0}}{v_\infty}\sqrt{\frac{v_\infty x}{\nu}} \tag{19.3-28a}$$

in which v_{y0} is the fluid velocity in the y-direction at the wall. This definition is applicable even when N_{A0} and N_{B0} are both nonzero. Note that v_{y0} varies as $1/\sqrt{x}$; this prediction, of course, is valid only in the region $v_\infty x/\nu \gg 1$ for which boundary-layer theory is valid.

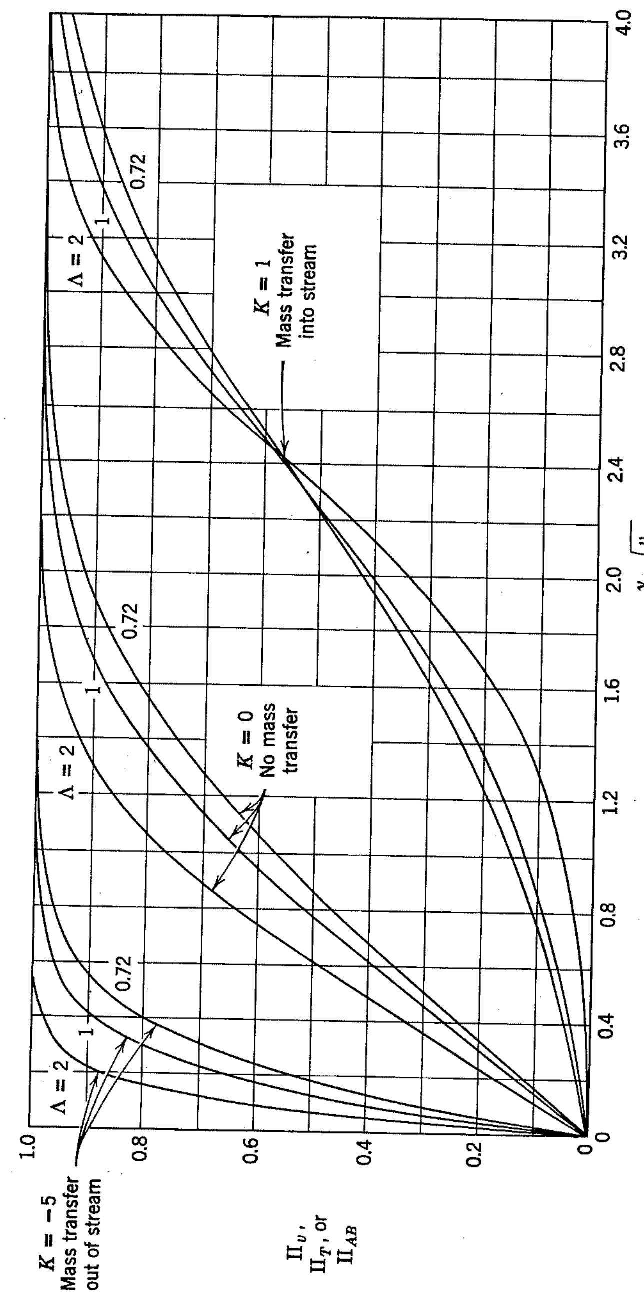


Fig. 19.3-2. Theoretical velocity, temperature, and composition profiles in the laminar boundary layer on a flat plate with mass transfer at the wall. [H. S. Mickley, R. C. Ross, A. L. Squyers, and W. E. Stewart, *NACA Technical Note* **3208** (1954).]

TABLE 19.3-1

DIMENSIONLESS GRADIENTS OF VELOCITY, TEMPERATURE, AND COMPOSITION CALCULATED FROM LAMINAR BOUNDARY-LAYER THEORY[a]

Dimensionless Gradient $2\Pi'(0, \Lambda, K)$ at the Flow Boundary

K	$\Lambda = 0.6$	$\Lambda = 0.7$	$\Lambda = 0.8$	$\Lambda = 0.9$	$\Lambda = 1.0$	$\Lambda = 1.1$	$\Lambda = 1.4$	$\Lambda = 2.0$	$\Lambda = 5.0$
1.0[e]	0.2321	0.2067	0.1831	0.1615	0.1419	0.1242	0.08219	0.03423	0.0002818
0.75[e]	0.4448	0.4290	0.4116	0.3933	0.3747	0.3560	0.3019	0.2098	0.02531
0.5[e]	0.6600	0.6644	0.6649	0.6626	0.6580	0.6516	0.6527	0.5587	0.2533
0.25[e]					0.9787				
0.0[c]	1.108	1.170	1.228	1.280	1.328[b]	1.374		1.689	
−0.5[d]	1.579	1.716	1.845	1.970	2.092	2.213		3.193	
−1.0[d]	2.069	2.288	2.501	2.709	2.916	3.121		4.892	
−1.5[d]	2.576	2.885	3.189	3.487	3.784	4.080		6.699	
−3.0[d]	4.174	4.765	5.354	5.943	6.529	7.115		12.41	
−5.0[d]	6.417	7.402	8.388	9.374	10.36	11.35		20.26	

[a] Based largely on the velocity profiles given by H. Schlichting and K. Bussmann, Schrift. d. deutsch. Akad. d. Luftfahrtforschung, *7B*, Heft 2 (1943); this table is reproduced from H. S. Mickley, R. C. Ross, A. L. Squyers, and W. E. Stewart, NACA TN 3208 (1954).
[b] H. Blasius, *Z. angew. Math. Phys.*, **56**, 1, 1–37 (1908).
[c] E. Pohlhausen, *Z. angew. Math. Mech.*, **1**, 115–121 (1921).
[d] J. A. Feyk, S.M. Thesis, Massachusetts Institute of Technology (1949).
[e] R. C. Ross, Private Communication (1950).

The fluxes of momentum, energy, and mass at the wall are given by the dimensionless expressions

$$\frac{\tau_{yx}|_{y=0}}{\rho v_\infty(0 - v_\infty)} = \frac{\Pi'(0, 1, K)}{2}\sqrt{\frac{\nu}{v_\infty x}} \tag{19.3–34}$$

$$\frac{q_y|_{y=0}}{\rho \hat{C}_p v_\infty(T_0 - T_\infty)} = \frac{\Pi'(0, \Lambda_T, K)}{2\Lambda_T}\sqrt{\frac{\nu}{v_\infty x}} \tag{19.3–35}$$

$$\frac{J_{Ay}^\star|_{y=0}}{c v_\infty(x_{A0} - x_{A\infty})} = \frac{\Pi'(0, \Lambda_{AB}, K)}{2\Lambda_{AB}}\sqrt{\frac{\nu}{v_\infty x}} \tag{19.3–36}$$

and the tabulated values of $\Pi'(0, \Lambda, K)$. We can thus compute the foregoing fluxes in terms of the dimensionless mass-transfer rate K. The solution for mass-transfer rate in terms of x_{A0} and $x_{A\infty}$ is illustrated in Example 19.3–1.

In a few limiting cases the profiles Π_T and Π_{AB} can be obtained analytically. For example, at small mass-transfer rates K and large values of Λ_T (the Prandtl number) or Λ_{AB} (the Schmidt number) the thermal or diffusional boundary layer is much thinner than the velocity boundary layer, and one needs to know only the velocity gradient at the wall to solve for the complete temperature and concentration profiles $\Pi(\eta, \Lambda, K)$. As we shall see, a small change in K has a marked effect on the temperature or concentration profile when Λ is large.

For small mass-transfer rates ($|K| \ll 1$), the velocity profile near the wall is

$$\Pi_v = \left(\frac{1.328}{2}\right)\eta \tag{19.3–37}$$

Here we have inserted the velocity gradient for $\eta = 0$ and $K = 0$, as given in Table 19.3–1. Application of Eqs. 19.3–27 and 32 then gives[5] for small K and $\Lambda \gg 1$

$$\Pi(\eta, \Lambda, K) \simeq \frac{\int_0^\eta \exp(\Lambda K\eta - 1.328\Lambda\eta^3/3!)\,d\eta}{\int_0^\infty \exp(\Lambda K\eta - 1.328\Lambda\eta^3/3!)\,d\eta} \tag{19.3–38}$$

and from Eq. 19.3–33 the corresponding temperature or concentration gradient at the wall is[6]

$$\Pi'(0, \Lambda, K) \simeq \frac{1}{\int_0^\infty \exp(\Lambda K\eta - 1.328\Lambda\eta^3/3!)\,d\eta} \tag{19.3–39}$$

[5] We use the symbol $\simeq$ to denote an asymptotic equality for large values of Λ.
[6] W. E. Stewart, Sc.D.Thesis, Massachusetts Institute of Technology (1951). See also, H. S. Mickley, R. C. Ross, A. L. Squyers, and W. E. Stewart, NACA TN 3208 (1954).

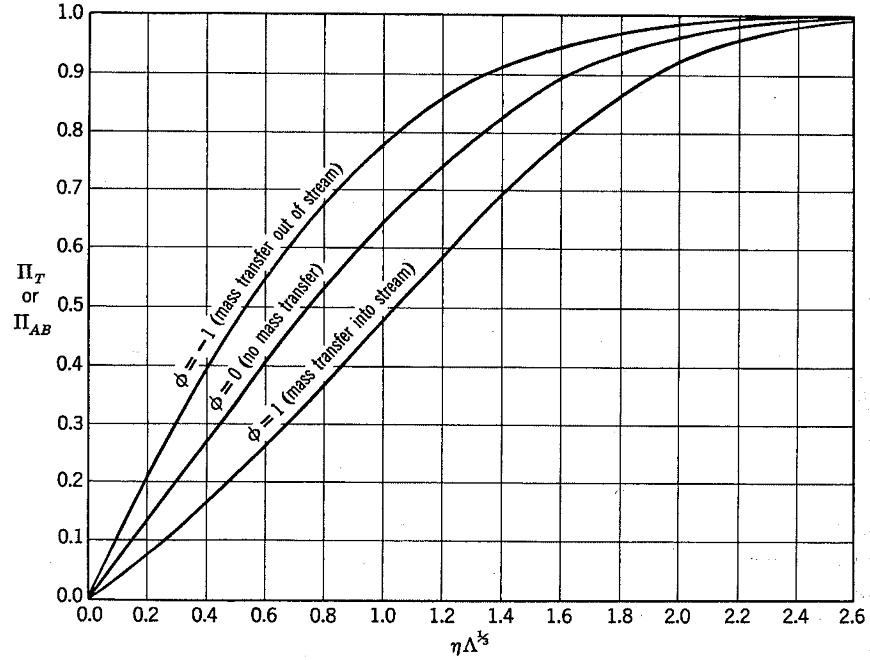


Fig. 19.3–3. Temperature or concentration profiles for high Prandtl or Schmidt number and various mass-transfer rates. The velocity profile used in these solutions is given by the line for $\Lambda = 1$ and $K = 0$ in Fig. 19.3–2.

In the following paragraphs we discuss the evaluation of the profiles and the gradients for small mass-transfer rates, K.

In the limit as $K \to 0$, the profiles take the form

$$\lim_{K\to 0} \Pi(\eta, \Lambda, K) \simeq \frac{\int_0^{\eta} \exp(-1.328\Lambda\eta^3/3!)\,d\eta}{\int_0^{\infty} \exp(-1.328\Lambda\eta^3/3!)\,d\eta} \tag{19.3–40}$$

$$= \frac{\Gamma(\frac{1}{3}, u)}{\Gamma(\frac{1}{3}, \infty)} \tag{19.3–41}$$

where

$$u = 1.328\frac{\Lambda\eta^3}{3!} \tag{19.3–42}$$

and $\Gamma(x, u)$ is the incomplete gamma function.[7] This profile is plotted in Fig. 19.3–3; note that it has nearly the same shape as the profiles for finite

[7] E. Jahnke and F. Emde, *Tables of Functions*, Dover, New York (1945), Fourth Edition, p. 22. The function $\Gamma(x, \infty) = \Gamma(x)$ is the complete gamma function.

Λ and $K = 0$ in Fig. 19.3–2. From Eq. 19.3–39 the slope of the profile at the wall is

$$\lim_{K\to 0} \Pi'(0, \Lambda, K) \simeq \frac{1}{\int_0^{\infty} \exp(-1.328\Lambda\eta^3/3!)\,d\eta}$$

$$= \left[\frac{1}{3}\left(\frac{3!}{1.328\Lambda}\right)^{1/3}\Gamma\left(\frac{1}{3}\right)\right]^{-1}$$

$$\doteq 0.6774\Lambda^{1/3} \tag{19.3–43}$$

This equation deviates by only 1 per cent, at $\Lambda = 2$, from the result obtained

TABLE 19.3–2

COMPARISON OF RESULTS FOR THE PROFILE GRADIENTS AT NEGLIGIBLE MASS-TRANSFER RATES

Λ	Predicted Values of $\Pi'(0, \Lambda, 0)$		
	Exact[a]	Asymptotic Solution $\Pi'(0, \Lambda, 0) = 0.6774\Lambda^{1/3}$	Pohlhausen Approximation $\Pi'(0, \Lambda, 0) = 0.664\Lambda^{1/3}$
0.6	0.552	0.571	0.560
0.8	0.614	0.629	0.616
1.0	0.664	0.677	0.664
2.0	0.845	0.853	0.837
7.0	1.29	1.296	1.27
10.0	1.46	1.459	1.43
15.0	1.67	1.671	1.64

[a] From the sources listed in Table 19.3–1.

by using the complete velocity profile, and the agreement becomes exact as $\Lambda \to \infty$. A detailed comparison is given in Table 19.3–2. Also shown are the results given by the equation

$$\Pi'(0, \Lambda, 0) \doteq 0.664\Lambda^{1/3} \tag{19.3–44}$$

which was given by Pohlhausen[8] as a curve-fit of the exact values and is good within ± 2 per cent down to $\Lambda = 0.6$. Insertion of Pohlhausen's approximation in Eqs. 19.2–34, 35, and 36 gives the following convenient analogies for low mass-transfer rates:

$$\frac{\tau_{yx}|_{y=0}}{\rho v_\infty(0 - v_\infty)} = \frac{q_y|_{y=0}}{\rho \hat{C}_p v_\infty (T_0 - T_\infty)}(\mathrm{Pr})^{2/3} = \frac{J^{\star}_{Ay}|_{y=0}}{c v_\infty (x_{A0} - x_{A\infty})}(\mathrm{Sc})^{2/3}$$
$$= 0.332\,\mathrm{Re}^{-1/2} \tag{19.3–45}$$

[8] E. Pohlhausen, *Z. angew. Math. Mech.*, **1**, 115–121 (1921).

in which $\text{Re} = v_\infty x/\nu$. This result justifies the Chilton-Colburn analogies (§§13.2 and 21.2) for this flow system for Pr or Sc greater than about $\frac{1}{2}$.

To extend the analytic solution to somewhat higher mass-transfer rates, the profiles are obtained from Eq. 19.3–38 by expanding exp $(\Lambda K\eta)$ in series and integrating term by term. The solution for any $K \ll 1$ is

$$\Pi(\eta, \Lambda, K) \simeq \frac{\sum_{m=0}^{\infty} c_m \phi^m \Gamma\left(\frac{m+1}{3}, u\right) \Big/ \Gamma\left(\frac{m+1}{3}, \infty\right)}{\sum_{m=0}^{\infty} c_m \phi^m} \tag{19.3–46}$$

Here $u = 1.328\Lambda\eta^3/3!$ as before, and the parameter ϕ is a dimensionless transfer rate:

$$\phi = \frac{K\Lambda}{\Pi'(0, \Lambda, 0)} \tag{19.3–47}$$

which, in view of Eq. 19.3–43 for large Λ, becomes

$$\phi \doteq \frac{K\Lambda^{2/3}}{0.6774} \tag{19.3–48}$$

The coefficients c_m are given by

$$c_m = \frac{1}{3}\frac{1}{m!}\Gamma\left(\frac{m+1}{3}\right)[\Gamma(\tfrac{4}{3})]^{-m-1} \tag{19.3–49}$$

TABLE 19.3–3

THE EFFECT OF MASS TRANSFER ON PROFILE GRADIENTS AT HIGH PRANDTL OR SCHMIDT NUMBER[a]

Mass flux Parameter $\lvert\phi\rvert$	$\frac{\Pi'(0, \Lambda, K)}{\Pi'(0, \Lambda, 0)}$ for positive ϕ (mass transfer into stream)	$\frac{\Pi'(0, \Lambda, K)}{\Pi'(0, \Lambda, 0)}$ for negative ϕ (mass transfer out of stream)
0.01	0.994	1.006
0.02	0.989	1.011
0.05	0.972	1.029
0.1	0.944	1.057
0.2	0.890	1.117
0.5	0.739	1.304
1.0	0.524	1.646
2.0	0.2300	2.421
3.0	0.0830	3.282
5.0	0.00613	5.140
∞	0	∞

[a] The values for $\lvert\phi\rvert < 5$ are taken from W. E. Stewart, Sc.D. Thesis, Massachusetts Institute of Technology (1951).

Some profiles computed from Eq. 19.3–46 are plotted in Fig. 19.3–3. The profiles are quite similar to those of Fig. 19.3–2 and show similar variations with mass-transfer rate. Note that the parameter ϕ, which determines the profile shapes, can have a large value in spite of the restriction $K \ll 1$ because Λ is assumed to be large. (See Eq. 19.3–48.) The gradients of these profiles at the wall are given by[6]

$$\frac{\Pi'(0, \Lambda, K)}{\Pi'(0, \Lambda, 0)} = \left(\sum_{m=0}^{\infty} c_m \phi^m\right)^{-1} \tag{19.3–50}$$

The ratio of gradients appearing in this equation gives a convenient measure of the effect of mass transfer on interphase transfer rates. Some values computed from this equation are given in Table 19.3–3; the results are further interpreted and applied to various mass-transfer problems in §21.7.

Example 19.3–1. Calculation of Mass-Transfer Rate

Give an expression for the local rate of sublimation from the plate shown in Fig. 19.3–1 into the laminar boundary layer. The mole fraction A next to the wall is $x_{A0} = 0.9$, the mole fraction A in the approaching stream is $x_{A\infty} = 0.01$, and the Schmidt number of the gas is constant at 2.0. M_A and M_B are equal, and the physical properties are constant.

Solution. Insertion of known quantities into Eq. 19.3–28 gives

$$\frac{2K}{\Pi_{AB}'(0, 2, K)} = \frac{0.9 - 0.01}{1 - 0.9} = 8.9 \tag{19.3–51}$$

To solve this equation for K, we evaluate the left-hand term as a function of K from Table 19.3–1:

K	$2\Pi'(0, 2, K)$	$2K/\Pi'(0, 2, K)$
1.00	0.03423	116.9
0.75	0.2098	14.30
0.50	0.5587	3.580
0.00	1.689	0.0000

From a plot of these values, we find that when $2K/\Pi'(0, 2, K) = 8.9$, $K = 0.65$. Therefore, the local mass flux is given by

$$K = \frac{2n_{A0}}{\rho v_\infty}\sqrt{\frac{v_\infty x \rho}{\mu}} = 0.65 \tag{19.3–52}$$

or

$$n_{A0} = 0.33\sqrt{\frac{\rho\mu v_\infty}{x}} \tag{19.3–53}$$

in the region $v_\infty x/\nu \gg 1$. In §21.7 this method of solution is simplified and extended to a variety of heat and mass-transfer problems.